Approximate Domain Unlearning for Vision-Language Models

immediate

Abstract

Vision-Language Models (VLMs) pre-trained on large-scale data learn rich representations that generalize across visual domains, yet this broad knowledge can become a liability when regulatory or privacy requirements demand the selective removal of domain-specific information. Existing machine unlearning methods operate at the class or sample level and cannot disentangle domain-level knowledge that shares the same semantic categories. We introduce **Approximate Domain Unlearning (ADU)**, a new problem formulation and method for selectively forgetting target domains from a prompted VLM while preserving accuracy on retained domains. ADU comprises two key components: (1) a **Domain Disentangling Loss (DDL)** that combines domain-classification cross-entropy with inter-domain Maximum Mean Discrepancy maximization to separate entangled domain representations, and (2) an **Instance-wise Prompt Generator (InstaPG)** that conditions deep visual prompts on individual image patch features via cross-attention, enabling fine-grained, input-adaptive forgetting. Experiments on Office-Home, Mini-DomainNet, and an ImageNet domain-unlearning benchmark show that ADU consistently outperforms prior approximate unlearning and prompt-tuning baselines, improving the harmonic mean of memorization and forgetting metrics by up to 16 points.

1 Introduction

Vision-Language Models (VLMs) such as CLIP [Radford et al., 2021], ALIGN [Jia et al., 2021], and CoCa [Yu et al., 2022] have demonstrated remarkable zero-shot generalization across diverse visual domains by leveraging contrastive pre-training on web-scale image-text pairs [Bommasani et al., 2021]. Their shared vision-language embedding space enables flexible transfer to downstream tasks through prompt tuning [Zhou et al., 2022b, Khattak et al., 2023] or lightweight adaptation [Kumar et al., 2022], making them attractive foundation models for real-world deployment.

However, this broad generalization introduces a critical challenge: once a VLM has absorbed knowledge from a particular visual domain, selectively removing that knowledge is non-trivial. Practical scenarios frequently demand such removal—for instance, a model trained on medical, satellite, and consumer imagery may need to forget the medical domain entirely to comply with data protection regulations [Fredrikson et al., 2015, Shokri et al., 2017], or an autonomous driving model may need to unlearn a geographically restricted domain after a licensing agreement expires. These requirements call for *domain-level* unlearning, a granularity that existing machine unlearning methods do not address.

Current machine unlearning approaches operate predominantly at the class level [Graves et al., 2021, Tarun et al., 2023, Fan et al., 2024] or the individual sample level [Cao and Yang, 2015, Bourtoule et al., 2021]. While effective for removing knowledge of specific categories or data points, these methods fundamentally assume that the knowledge to be forgotten is semantically distinct from the knowledge to be retained. Domain unlearning breaks this assumption: the *same* object classes

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appear across both the forget and retain domains, differing only in visual style, texture, or acquisition conditions (see Figure 1).

The core technical challenge lies in the *entanglement* of domain features within the VLM's representation space. Pre-trained VLMs encode domain-variant and domain-invariant features in a shared embedding, making it difficult to isolate and remove domain-specific information without degrading performance on retained domains. Our analysis confirms this: domain classification accuracy in the pre-trained CLIP feature space is only 25.80%, indicating that domain representations are heavily overlapping.

To address these challenges, we propose **Approximate Domain Unlearning (ADU)**, comprising two complementary components. First, we introduce a **Domain Disentangling Loss (DDL)** that encourages the model to separate domain-specific features by jointly optimizing a domain-classification cross-entropy loss and maximizing the Maximum Mean Discrepancy (MMD) [Gretton et al., 2012] between domain distributions in the feature space. This explicit disentanglement raises domain separability from 25.80% to 79.43%, enabling targeted forgetting. Second, we propose an **Instance-wise Prompt Generator (InstaPG)** that augments deep visual prompts with input-dependent information through a cross-attention mechanism conditioned on image patch features.

Our contributions are summarized as follows:

- We formalize the **Approximate Domain Unlearning** problem for VLMs, where the same semantic categories appear in both forget and retain domains, making class-level unlearning insufficient.
- We propose the **Domain Disentangling Loss (DDL)**, combining domain-classification cross-entropy with MMD maximization to explicitly separate entangled domain representations.
- We introduce the **Instance-wise Prompt Generator (InstaPG)**, a cross-attention module that conditions deep visual prompts on individual image patch features, enabling fine-grained, input-adaptive domain forgetting.
- We conduct extensive experiments on three benchmarks, demonstrating that ADU consistently outperforms existing baselines across varying numbers of forget domains.

2 Related Work

Machine Unlearning. Machine unlearning aims to remove the influence of specific training data from a learned model without full retraining [Cao and Yang, 2015, Xu et al., 2024]. Exact unlearning methods guarantee complete removal through data partitioning and selective retraining [Bourtoule et al., 2021, Ginart et al., 2019] but incur prohibitive computational costs for large models. Approximate unlearning methods trade exactness for efficiency: influence-function-based approaches [Izzo et al., 2021, Golatkar et al., 2020a,b] estimate parameter updates, while fine-tuning-based methods [Golatkar et al., 2021, Graves et al., 2021, Warnecke et al., 2023] modify the model through additional training on retain data or noised forget data. Recent work has explored class-level forgetting through gradient-based saliency [Fan et al., 2024], boundary manipulation [Chen et al., 2023], and knowledge distillation [Chundawat et al., 2023, Tarun et al., 2023]. However, these methods uniformly assume that the forget set is defined at the class or sample level. Our work instead targets domain-level forgetting, where the same classes appear in both forget and retain sets.

Vision-Language Models and Prompt Tuning. Large-scale VLMs [Radford et al., 2021, Jia et al., 2021, Li et al., 2022, 2023] learn aligned vision-language representations through contrastive or generative pre-training on web-scale data. Adapting these models to downstream tasks via full fine-tuning is expensive and risks catastrophic forgetting [Kumar et al., 2022]. Prompt tuning offers a parameter-efficient alternative: CoOp [Zhou et al., 2022b] learns continuous text prompts, MaPLe [Khattak et al., 2023] extends this to multi-modal deep prompting, and domain-specific prompt methods [Zhou et al., 2022a] condition prompts on domain information. In the unlearning context, recent methods have applied prompt-based adaptation for VLM forgetting [Li et al., 2024] and explored black-box approaches [Kuwana et al., 2024]. LP++ [Huang et al., 2024] adapts VLMs through logit-level post-processing. Our InstaPG differs from these approaches by generating instance-adaptive prompts through cross-attention over image patch features, providing fine-grained control over per-input forgetting behavior.

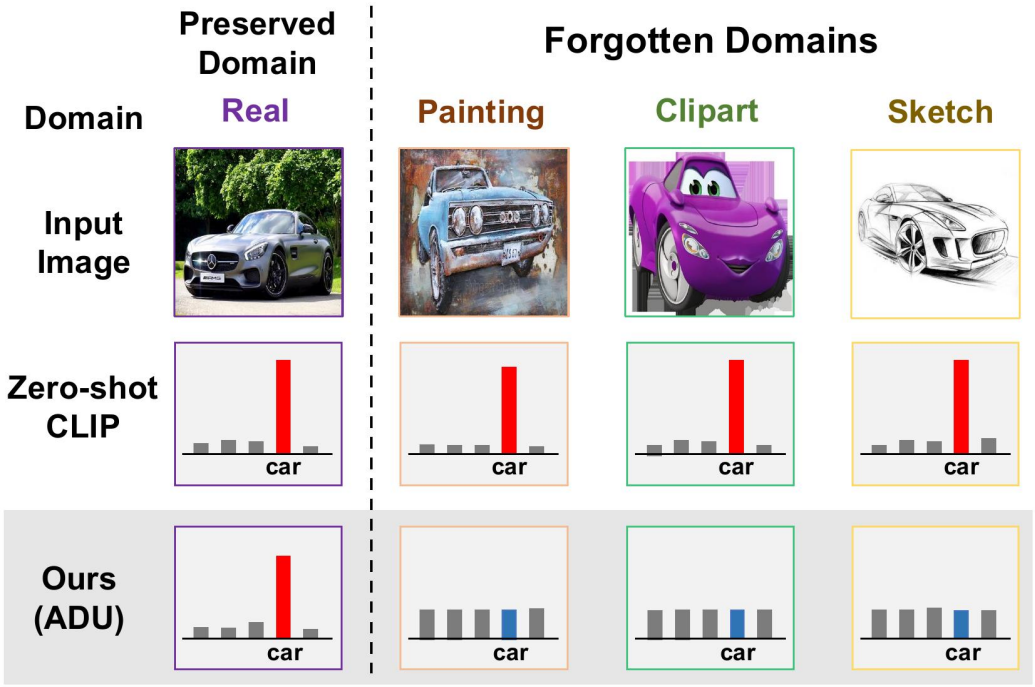


Figure 1: **Illustration of Approximate Domain Unlearning (ADU).** Given a pre-trained VLM with knowledge of multiple visual domains, ADU selectively forgets specified domains while retaining performance on the remaining domains. The same semantic classes appear across both forgotten and retained domains, rendering class-level unlearning insufficient.

Domain Adaptation and Distribution Shift. Domain adaptation methods seek to transfer knowledge across domains with differing distributions [Ganin et al., 2016, Long et al., 2015, 2017]. A central technique is minimizing distribution discrepancy, often measured by MMD [Gretton et al., 2012] or adversarial domain discriminators [Ganin et al., 2016]. While domain adaptation aims to *align* distributions for positive transfer, our work pursues the opposite: *separating* domain distributions. Our DDL repurposes MMD maximization combined with domain classification to disentangle domain features, enabling selective forgetting.

3 Method

In this section, we introduce our framework for Approximate Domain Unlearning (ADU) in vision-language models. We first formalize the problem setup (§3.1), then describe the baseline memorize-forget objective (§3.2), followed by our Domain Disentangling Loss (§3.3) and Instance-wise Prompt Generator (§3.4). An overview of the full method is illustrated in Figure 1.

3.1 Problem Formulation

Let $\mathcal{D} = \{\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_K\}$ denote a collection of K visual domains, where each domain $\mathcal{D}_k$ contains image-label pairs $\{(\mathbf{x}_i^{(k)}, y_i^{(k)})\}$ sharing a common set of C semantic categories but differing in visual style (*e.g.*, Art, Clipart, Product, Real World in Office-Home). Given a pre-trained vision-language model f_θ (*e.g.*, CLIP [Radford et al., 2021]) that performs well across all domains, a domain unlearning request specifies a forget set $\mathcal{D}_{\text{forget}} \subset \mathcal{D}$ whose knowledge should be erased, while the complementary retain set $\mathcal{D}_{\text{retain}} = \mathcal{D} \setminus \mathcal{D}_{\text{forget}}$ should be preserved.

Formally, the objective of approximate domain unlearning is to obtain a model $f_{\theta'}$ that simultaneously satisfies:

$$\min_{\theta'} \underbrace{\sum_{\mathcal{D}_k \in \mathcal{D}_{\text{retain}}} \mathcal{L}_{\text{task}}(f_{\theta'}, \mathcal{D}_k)}_{\text{maintain retain-domain performance}} \quad \text{while} \quad \underbrace{\max_{\theta'} \; H\big(f_{\theta'}(\mathbf{x})\big), \; \forall \mathbf{x} \in \mathcal{D}_{\text{forget}},}_{\text{erase forget-domain knowledge}} \tag{1}$$

where $\mathcal{L}_{\text{task}}$ is the task loss (*e.g.*, cross-entropy for classification) and $H(\cdot)$ denotes the entropy of the predicted distribution. We evaluate unlearning quality using three metrics: **Mem** (accuracy on $\mathcal{D}_{\text{retain}}$), **For** (error rate on $\mathcal{D}_{\text{forget}}$; higher is better), and the harmonic mean $H = 2 \cdot \text{Mem} \cdot \text{For}/(\text{Mem} + \text{For})$.

To avoid modifying the pre-trained backbone and to enable parameter-efficient adaptation, we adopt deep visual prompting [Dosovitskiy et al., 2021]. Specifically, we prepend a set of N_p learnable prompt tokens $\{\mathbf{p}_l\}_{l=1}^{N_p}$ to the patch token sequence at each layer of the vision transformer encoder. Only these prompt tokens and a lightweight domain classifier head are trained; the backbone parameters of CLIP remain frozen.

3.2 Baseline: Memorize-Forget Losses

Our baseline approach employs two complementary losses applied to the classification logits produced by the prompted CLIP model.

Memorize Loss. For samples from the retain domains, we minimize the standard cross-entropy loss to preserve classification accuracy:

$$\mathcal{L}_{\text{memorize}} = \frac{1}{|\mathcal{B}_{\text{retain}}|} \sum_{(\mathbf{x},y) \in \mathcal{B}_{\text{retain}}} \mathcal{L}_{\text{CE}}\big(f_{\theta'}(\mathbf{x}),\, y\big), \tag{2}$$

where $\mathcal{B}_{\text{retain}}$ denotes the retain-domain samples in the current mini-batch.

Forget Loss. For samples from the forget domains, we maximize the entropy of the output distribution to erase discriminative knowledge [Graves et al., 2021, Chundawat et al., 2023]:

$$\mathcal{L}_{\text{forget}} = -\frac{1}{|\mathcal{B}_{\text{forget}}|} \sum_{\mathbf{x} \in \mathcal{B}_{\text{forget}}} H\big(f_{\theta'}(\mathbf{x})\big), \quad \text{where} \quad H(\mathbf{p}) = -\sum_{c=1}^{C} p_c \log p_c. \tag{3}$$

Since we want to *maximize* entropy, minimizing $\mathcal{L}_{\text{forget}}$ drives predictions on forgotten domains toward the uniform distribution. The baseline training objective combines these two losses:

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{memorize}} + \lambda_f \, \mathcal{L}_{\text{forget}}, \tag{4}$$

where λ_f controls the strength of forgetting. While effective in simple cases, this baseline struggles when domain-specific and domain-general features are entangled in the shared representation space, because suppressing forget-domain predictions may inadvertently degrade retain-domain performance.

3.3 Domain Disentangling Loss (DDL)

A key challenge in domain unlearning is that features learned by the vision encoder often entangle domain-specific style information with domain-invariant semantic content. When these representations overlap, entropy maximization on forget domains can corrupt features that are also useful for retain domains. To address this, we introduce a *Domain Disentangling Loss* (DDL) that explicitly encourages the model to learn separable, domain-aware representations. DDL consists of two complementary components: a domain classification loss that promotes domain-discriminative features, and an MMD term that pushes inter-domain feature distributions apart.

Domain Classification Loss. We attach a linear domain classifier g_ϕ on top of the image features $\mathbf{z} = f_{\theta'}(\mathbf{x})$ and train it to predict the domain identity. For a dataset with K domains and C classes, the classifier produces logits over $K \times C$ domain-class combinations:

$$\mathcal{L}_{\text{CE}}^{\text{dom}} = \frac{1}{|\mathcal{B}|} \sum_{(\mathbf{x},d) \in \mathcal{B}} \mathcal{L}_{\text{CE}}\big(g_\phi(\mathbf{z}),\, d\big), \tag{5}$$

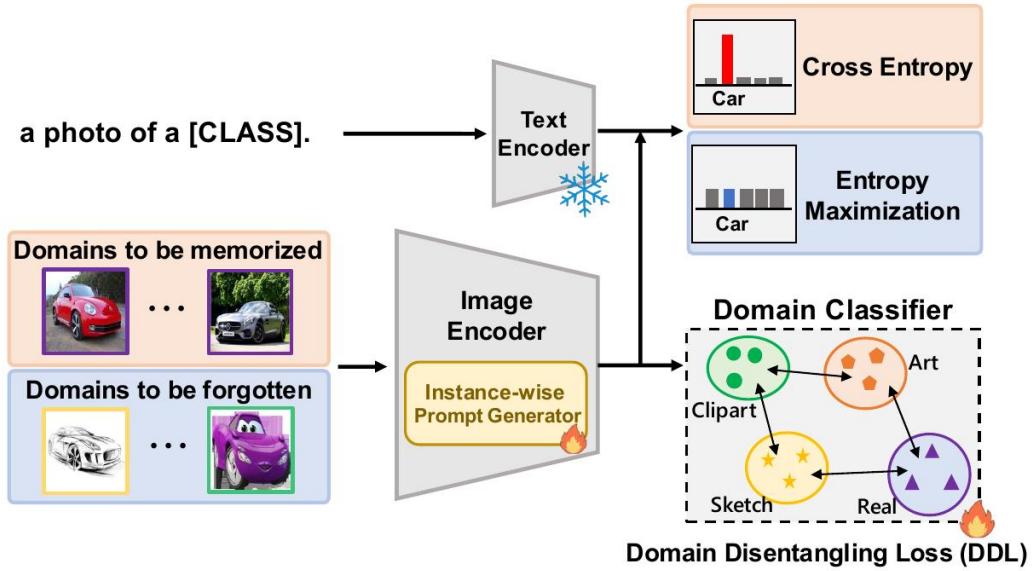


Figure 2: **Overview of the Proposed ADU Framework.** The frozen CLIP ViT-B/16 backbone is augmented with deep vision prompts. The Domain Disentangling Loss (DDL) explicitly separates domain distributions via domain-classification cross-entropy and inter-domain MMD maximization. The Instance-wise Prompt Generator (InstaPG) conditions the prompts on image patch features via cross-attention.

where $d \in \{1, \ldots, K\}$ is the domain label. Minimizing this loss encourages the representation to encode domain-discriminative information, which is essential for selectively unlearning specific domains while retaining others [Ganin et al., 2016].

MMD Maximization. While the domain classifier encourages domain awareness, it does not explicitly enforce geometric separation between domain distributions in feature space. We therefore incorporate a Maximum Mean Discrepancy (MMD) [Gretton et al., 2012] term. Given features from two domains $\mathcal{D}_i$ and $\mathcal{D}_j$, the MMD with a Gaussian kernel $k(\cdot, \cdot)$ is:

$$\mathrm{MMD}^2(\mathcal{D}_i, \mathcal{D}_j) = \mathbb{E}[k(\mathbf{z}_i, \mathbf{z}_i')] + \mathbb{E}[k(\mathbf{z}_j, \mathbf{z}_j')] - 2\,\mathbb{E}[k(\mathbf{z}_i, \mathbf{z}_j)], \tag{6}$$

where $\mathbf{z}_i, \mathbf{z}_i' \sim \mathcal{D}_i$ and $\mathbf{z}_j, \mathbf{z}_j' \sim \mathcal{D}_j$. We compute the total pairwise MMD across all domain pairs present in each mini-batch:

$$\mathcal{L}_{\mathrm{MMD}} = \frac{1}{\binom{K'}{2}} \sum_{i<j} \mathrm{MMD}^2(\mathcal{D}_i, \mathcal{D}_j), \tag{7}$$

where K' is the number of distinct domains in the batch and the bandwidth σ is set via the median heuristic. Whereas domain adaptation *minimizes* MMD to align distributions [Long et al., 2015], we *maximize* $\mathcal{L}_{\mathrm{MMD}}$ to push domain distributions apart.

Combined DDL. The full Domain Disentangling Loss combines both terms:

$$\mathcal{L}_{\mathrm{DDL}} = \mathcal{L}_{\mathrm{CE}}^{\mathrm{dom}} - \gamma\, \mathcal{L}_{\mathrm{MMD}}, \tag{8}$$

where $\gamma > 0$ controls the weight of MMD maximization. By minimizing $\mathcal{L}_{\mathrm{DDL}}$, we jointly train the domain classifier and push domain distributions apart, yielding a feature space where domain-specific signals are well separated and can be selectively erased.

3.4 Instance-wise Prompt Generator (InstaPG)

The deep vision prompts described in §3.2 are shared across all input images, providing a single set of adaptation parameters regardless of the input. However, different images—even within the same

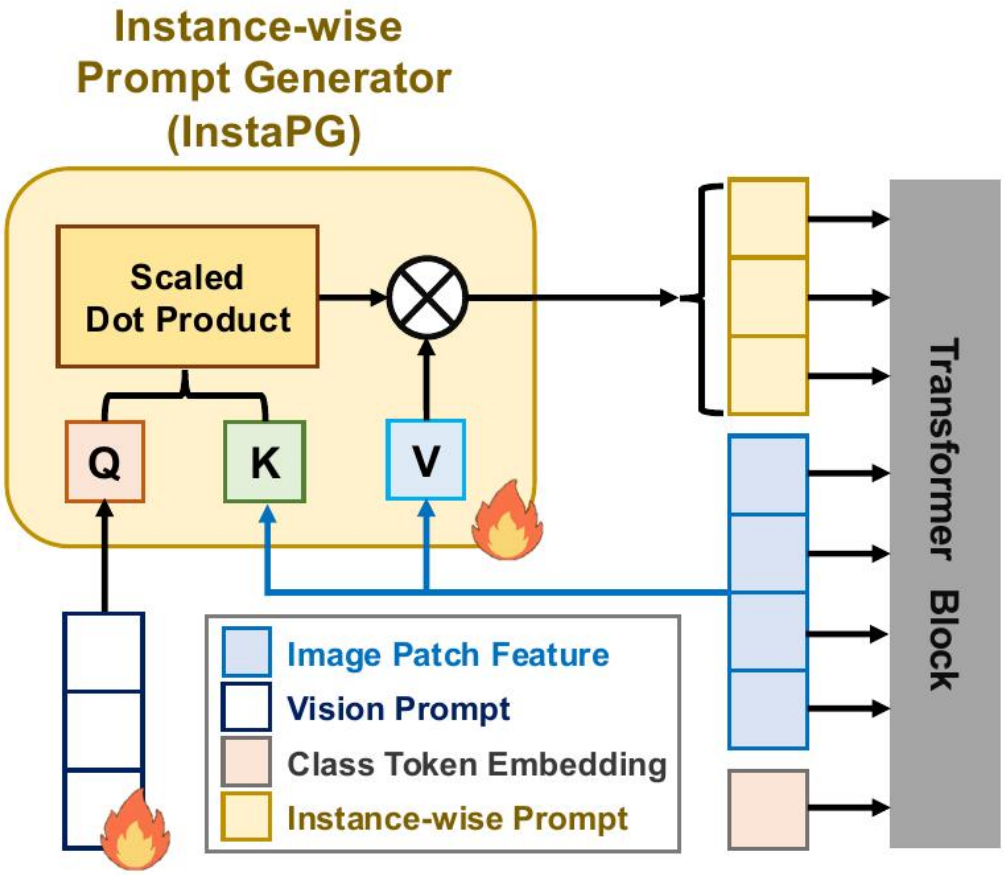


Figure 3: **Instance-wise Prompt Generator (InstaPG).** A learnable knowledge bank is conditioned on image patch features via cross-attention, producing instance-adaptive vision prompts that encode fine-grained, per-image domain cues.

domain—may require different degrees of domain-specific adaptation depending on their content. We propose an *Instance-wise Prompt Generator* (InstaPG) that dynamically conditions the vision prompts on each input image's patch features, producing instance-specific prompts that capture fine-grained domain cues.

Knowledge Bank. InstaPG maintains a learnable knowledge bank $\mathbf{K} \in \mathbb{R}^{L \times D}$, where L is the number of prompt embeddings and D is the embedding dimension.

Cross-Attention Conditioning. Given an input image $\mathbf{x}$, the vision encoder produces intermediate patch features $\mathbf{F} \in \mathbb{R}^{B \times P \times D}$ at a designated layer, where B is the batch size and P is the number of spatial patches. The knowledge bank is refined through cross-attention:

$$\hat{\mathbf{K}} = \mathrm{CrossAttn}(\mathbf{Q} = \mathbf{K},\ \mathbf{K} = \mathbf{F},\ \mathbf{V} = \mathbf{F}), \tag{9}$$

where the knowledge bank acts as queries and the image patch features serve as keys and values. Each cross-attention block consists of layer normalization, multi-head cross-attention, and a feed-forward network with GEGLU activation, connected via residual connections:

$$\mathbf{K}' = \mathrm{MHA}\big(\mathrm{LN}(\mathbf{K}),\ \mathrm{LN}(\mathbf{F})\big) + \mathbf{K}, \quad \hat{\mathbf{K}} = \mathrm{FFN}\big(\mathrm{LN}(\mathbf{K}')\big) + \mathbf{K}'. \tag{10}$$

Positional embeddings are added to the patch features before cross-attention to preserve spatial information.

Prompt Augmentation. The output $\hat{\mathbf{K}} \in \mathbb{R}^{B \times L \times D}$ provides instance-specific prompt tokens inserted at designated layers of the vision transformer, enabling instance-adaptive modulation of the visual representation. This allows the model to capture fine-grained, input-dependent domain signals that static prompts cannot represent—critical for precise domain-level unlearning.

3.5 Overall Training Objective

The full ADU training objective combines all loss components:

$$\mathcal{L} = \mathcal{L}_{\text{memorize}} + \lambda_f \, \mathcal{L}_{\text{forget}} + \lambda_d \, \mathcal{L}_{\text{DDL}}, \tag{11}$$

where λ_f and λ_d are hyperparameters controlling the relative importance of forgetting and domain disentangling, respectively. Expanding $\mathcal{L}_{\text{DDL}}$:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{memorize}}}_{\text{retain accuracy}} + \lambda_f \underbrace{\mathcal{L}_{\text{forget}}}_{\text{erase knowledge}} + \lambda_d \underbrace{\mathcal{L}_{\text{CE}}^{\text{dom}}}_{\text{domain classification}} - \lambda_d \gamma \underbrace{\mathcal{L}_{\text{MMD}}}_{\text{distribution separation}}. \tag{12}$$

During training, only the vision prompt tokens, the InstaPG module (knowledge bank and cross-attention layers), the text prompt tokens, and the domain classifier head are optimized. The CLIP backbone remains entirely frozen, ensuring parameter efficiency and preventing catastrophic forgetting of the pre-trained representations.

4 Experiments

4.1 Setup

Datasets. We evaluate our method on three multi-domain benchmarks. (1) **ImageNet** [Deng et al., 2009]: We use real photographs from ImageNet and sketches from ImageNet-Sketch [Wang et al., 2019] as two domains, testing only $|\mathcal{D}_{\text{forget}}| = 1$. (2) **Office-Home** [Venkateswara et al., 2017]: This dataset contains images from four domains—Art, Clipart, Product, and Real—spanning 65 categories. We evaluate $|\mathcal{D}_{\text{forget}}| \in \{1, 2, 3\}$. (3) **Mini DomainNet** [Zhou et al., 2021]: A subset of DomainNet [Peng et al., 2019] with four domains—Clipart, Painting, Real, and Sketch. We evaluate $|\mathcal{D}_{\text{forget}}| \in \{1, 2, 3\}$. For each setting of $|\mathcal{D}_{\text{forget}}|$, we report the average performance over all possible combinations of forgotten and retained domains.

Metrics. We use three metrics: (i) *Mem*: accuracy on retained (memorized) domains, (ii) *For*: error rate on forgotten domains (higher means better forgetting), and (iii) H: the harmonic mean of *Mem* and *For*, which captures the trade-off between memorization and forgetting. Higher values indicate better performance for all metrics.

Baselines. We compare against three CLIP fine-tuning methods repurposed for domain unlearning: LP++ [Huang et al., 2024], CLIPFit [Li et al., 2024], and BBF [Kuwana et al., 2024]. We also include a *Baseline* that trains deep visual prompts using only $\mathcal{L}_{\text{memorize}}$ and $\mathcal{L}_{\text{forget}}$ without DDL or InstaPG.

Implementation Details. We use CLIP ViT-B/16 [Radford et al., 2021] as the backbone, keeping all pretrained parameters frozen. Deep visual prompts are inserted at multiple transformer layers. The Instance-wise Prompt Generator (InstaPG) uses cross-attention to condition prompts on image patch tokens. Training is performed in a few-shot setting with a fixed number of shots per domain.

4.2 Main Results

Table 1 presents the comparison of our method with baselines across all three datasets. On ImageNet with $|\mathcal{D}_{\text{forget}}| = 1$, our method achieves an H score of 77.02, outperforming the Baseline (74.66), CLIPFit (71.31), LP++ (50.69), and BBF (45.56). Our method attains both the highest *Mem* (64.13) and *For* (96.73), indicating effective forgetting without sacrificing retained-domain accuracy.

On Office-Home, the improvements are more pronounced. For $|\mathcal{D}_{\text{forget}}| = 1$, our method achieves $H = 69.96$ compared to the Baseline's 52.59, a gain of over 17 points. This advantage grows with more domains to forget: H improves from 54.19 to 73.58 for $|\mathcal{D}_{\text{forget}}| = 2$ and from 59.47 to 75.89 for

Table 1: **Comparison with Baseline CLIP Fine-tuning Methods.** The average performance of all possible combinations of the domains to be forgotten/retained are reported for various number of domains to be forgotten $|\mathcal{D}_{\text{forget}}| \in \{1, 2, 3\}$. ImageNet has only two domains, so only the case $|\mathcal{D}_{\text{forget}}| = 1$ is tested. Performance is evaluated using the three metrics: (i) *Mem*: the accuracy of the classes for data from memorized domains, (ii) *For*: the error of the classes for data from forgotten domains, and (iii) H: the harmonic mean of *Mem* and *For*. Higher values mean better performance. The *Baseline* in the table refers to the method that trains the vision prompt using $\mathcal{L}_{\text{memorize}}$ and $\mathcal{L}_{\text{forget}}$ in Sec. 3.2, representing a straightforward and conventional approach to unlearning.

$\lvert\mathcal{D}_{\text{forget}}\rvert$	Method	ImageNet			Office-Home			Mini DomainNet		
		$H\uparrow$	*Mem*↑	*For*↑	$H\uparrow$	*Mem*↑	*For*↑	$H\uparrow$	*Mem*↑	*For*↑
$\lvert\mathcal{D}_{\text{forget}}\rvert = 1$	LP++ [Huang et al., 2024]	50.69	62.22	42.76	30.46	**85.06**	18.55	31.73	84.72	19.52
	CLIPFit [Li et al., 2024]	71.31	63.04	82.13	43.44	83.20	29.40	53.56	84.32	39.24
	BBF [Kuwana et al., 2024]	45.56	35.92	65.91	31.25	80.94	19.74	32.12	81.80	20.03
	Baseline	74.66	61.23	96.43	52.59	79.96	39.88	62.07	**85.15**	49.40
	Ours	**77.02**	**64.13**	**96.73**	**69.96**	77.93	**64.34**	**75.56**	78.32	**73.06**
$\lvert\mathcal{D}_{\text{forget}}\rvert = 2$	LP++ [Huang et al., 2024]	×	×	×	31.11	**85.54**	19.01	32.21	**85.29**	19.85
	CLIPFit [Li et al., 2024]	×	×	×	40.53	83.64	26.74	51.35	84.74	36.83
	BBF [Kuwana et al., 2024]	×	×	×	32.54	80.42	20.54	47.95	61.48	40.52
	Baseline	×	×	×	54.19	80.23	41.23	61.97	84.71	48.96
	Ours	×	×	×	**73.58**	75.61	**71.89**	**77.03**	76.51	**77.66**
$\lvert\mathcal{D}_{\text{forget}}\rvert = 3$	LP++ [Huang et al., 2024]	×	×	×	33.57	**86.22**	20.84	34.73	**85.87**	21.77
	CLIPFit [Li et al., 2024]	×	×	×	50.02	85.68	35.32	52.88	85.39	38.30
	BBF [Kuwana et al., 2024]	×	×	×	39.60	78.13	26.80	39.29	80.29	26.05
	Baseline	×	×	×	59.47	80.96	48.79	68.82	84.25	58.27
	Ours	×	×	×	**75.89**	72.15	**80.77**	**79.00**	75.00	**83.94**

$|\mathcal{D}_{\text{forget}}| = 3$. Notably, the *For* metric increases substantially (e.g., 39.88 to 64.34 for $|\mathcal{D}_{\text{forget}}| = 1$), demonstrating that our approach enables much more effective domain forgetting.

On Mini DomainNet, H improves from 62.07 to 75.56, 61.97 to 77.03, and 68.82 to 79.00 for $|\mathcal{D}_{\text{forget}}| \in \{1, 2, 3\}$, with similarly large gains in *For* across all settings.

Overall, LP++, CLIPFit, and BBF achieve high *Mem* but fail to effectively forget target domains, while ADU substantially improves the memorization–forgetting balance.

4.3 Ablation Study

We conduct ablation studies to evaluate the contribution of each proposed component. Table 2 reports the results of adding DDL and InstaPG individually and jointly.

On Office-Home with $|\mathcal{D}_{\text{forget}}| = 1$, adding InstaPG alone improves H from 52.59 to 56.41, while adding DDL alone yields a larger improvement to 60.82. Combining both achieves the best performance at $H = 69.96$. A similar trend is observed on Mini DomainNet: the Baseline achieves $H = 62.07$, which increases to 64.06 with InstaPG, 74.23 with DDL, and 75.56 with both. DDL contributes more substantially to the H score by improving the *For* metric, while InstaPG provides complementary gains, particularly when combined with DDL. The synergy between both components is evident across all settings of $|\mathcal{D}_{\text{forget}}|$.

Table 3 further analyzes the two loss components within DDL: cross-entropy (CE) and Maximum Mean Discrepancy (MMD) [Gretton et al., 2012]. On Office-Home with $|\mathcal{D}_{\text{forget}}| = 1$, using MMD alone achieves $H = 68.62$, CE alone achieves $H = 64.01$, and combining both yields $H = 69.96$. MMD provides a stronger individual contribution, but the combination consistently performs best across all $|\mathcal{D}_{\text{forget}}|$ settings, confirming that CE and MMD offer complementary signals for domain disentanglement.

4.4 Domain Separability Analysis

To understand how our method improves domain-level control, we evaluate domain classification accuracy using a linear classifier trained on the learned feature representations. Table 4 reports results

Table 2: **Ablation Study.** The ablation results of Domain Disentangling Loss (DDL) and Instance-wise Prompt Generator (InstaPG) are reported. While our method achieves improvements with either DDL or InstaPG alone, combining both leads to even better balanced trade-off performance.

Dataset	DDL	InstaPG	$\lvert\mathcal{D}_{\text{forget}}\rvert = 1$			$\lvert\mathcal{D}_{\text{forget}}\rvert = 2$			$\lvert\mathcal{D}_{\text{forget}}\rvert = 3$		
			H ↑	*Mem*↑	*For*↑	*H* ↑	*Mem*↑	*For*↑	*H* ↑	*Mem*↑	*For*↑
Office-Home	-	-	52.59	79.96	39.88	54.19	80.23	41.23	59.47	80.96	48.79
	-	✓	56.41	**83.55**	44.05	61.01	**83.33**	48.60	70.60	**81.55**	63.28
	✓	-	60.82	74.51	51.72	60.01	76.22	49.94	63.12	77.74	54.57
	✓	✓	**69.96**	77.93	**64.34**	**73.58**	75.61	**71.89**	**75.89**	72.15	**80.77**
Mini DomainNet	-	-	62.07	85.15	49.40	61.97	84.71	48.96	68.82	**84.25**	58.27
	-	✓	64.06	**85.46**	51.64	65.92	**85.12**	53.96	74.17	83.68	66.68
	✓	-	74.23	78.00	70.87	75.28	77.57	73.20	77.60	77.42	77.90
	✓	✓	**75.56**	78.32	**73.06**	**77.03**	76.51	**77.66**	**81.78**	77.96	**86.12**

Table 3: **Ablation Studies of Loss Functions in Domain Disentangling Loss (DDL).** DDL employs two loss functions: Cross-Entropy (CE) and Maximum Mean Discrepancy (MMD). Using both losses together yields the best-balanced trade-off, as shown by H.

CE	MMD	$\lvert\mathcal{D}_{\text{forget}}\rvert = 1$			$\lvert\mathcal{D}_{\text{forget}}\rvert = 2$			$\lvert\mathcal{D}_{\text{forget}}\rvert = 3$		
		$H \uparrow$	$Mem \uparrow$	$For \uparrow$	$H \uparrow$	$Mem \uparrow$	$For \uparrow$	$H \uparrow$	$Mem \uparrow$	$For \uparrow$
-	-	56.41	**83.55**	44.05	61.01	**83.33**	48.60	70.60	81.55	63.28
-	✓	68.62	82.41	59.47	72.21	81.01	65.66	66.76	**82.38**	57.12
✓	-	64.01	82.97	53.62	69.33	80.73	61.24	71.53	81.43	64.55
✓	✓	**69.96**	77.93	**64.34**	**73.58**	75.61	**71.89**	**75.89**	72.15	**80.77**

on Office-Home when the Art domain is forgotten. Before unlearning, the domain classification accuracy is only 25.80%, indicating that CLIP's pretrained features heavily entangle domain information. The Baseline (without DDL and InstaPG) marginally improves this to 31.06%. In contrast, our full method achieves 79.43%, demonstrating that DDL and InstaPG effectively disentangle domain-specific features in the representation space.

Figure 4 provides a qualitative view via t-SNE visualizations when the Real domain is forgotten on Office-Home. In the zero-shot CLIP representation (left), samples from different domains are heavily intermixed, making domain-selective forgetting difficult. After applying our method (right), domains form distinct clusters, enabling the model to selectively suppress the forgotten domain while preserving others.

4.5 Sensitivity Analysis

Number of Training Samples. Figure 5 shows sensitivity to the number of training samples per domain. Our method consistently outperforms the Baseline regardless of shot count, maintaining a clear advantage even in low-shot regimes.

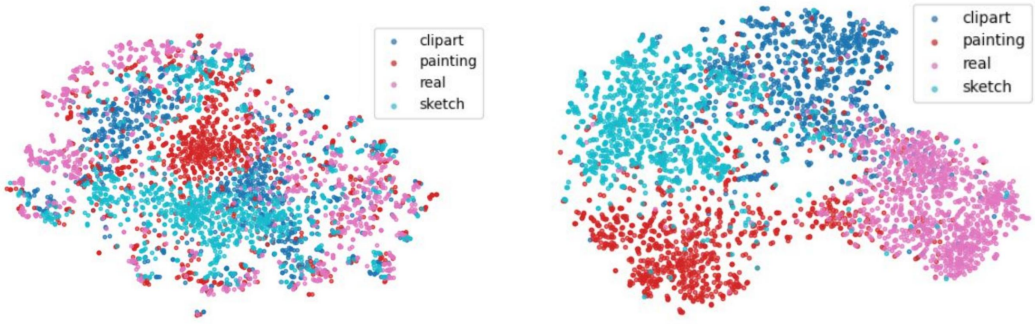
Hyperparameter Analysis. Figure 6 examines the impact of the loss weights γ and λ in DDL. Our method shows stable performance across a range of values for both hyperparameters, indicating that the method is not overly sensitive to these choices.

4.6 Qualitative Analysis

Figure 7 visualizes attention maps generated using GradCAM [Selvaraju et al., 2017]. The attention maps show that our method attends to domain-relevant visual cues, focusing on style and texture characteristics that distinguish domains. This confirms that the learned prompts capture meaningful

Table 4: **Domain Classification Accuracy when *Art* is Forgotten on Office-Home.**

Method	Domain Classification Accuracy [%]
Before Unlearning	25.80
Ours w/o DDL and InstaPG	31.06
Ours	79.43



(a) Zero-shot CLIP (b) Ours

Figure 4: **t-SNE Visualization, Where the Domain to Be Forgotten is *Real*.** (a) Zero-shot CLIP features show heavily entangled domains. (b) Our method produces well-separated domain clusters, enabling selective forgetting.

domain-specific information, enabling the model to discriminate between domains for selective forgetting.

5 Limitations

Our method introduces additional computational overhead through the Instance-wise Prompt Generator, which requires cross-attention computation for each input image. Furthermore, the approach assumes access to domain labels during training, which may not always be available in practice. The effectiveness of domain disentanglement may also degrade when domain boundaries are ambiguous or when domains share substantial visual overlap.

6 Conclusions

We introduced Approximate Domain Unlearning (ADU) for selectively forgetting domain-specific knowledge in VLMs while preserving retained-domain performance. ADU combines Domain Disentangling Loss (DDL), which separates entangled representations via cross-entropy and MMD maximization, with Instance-wise Prompt Generator (InstaPG), which adapts visual prompts per image via cross-attention on patch tokens. Experiments on ImageNet, Office-Home, and Mini DomainNet show that ADU consistently outperforms baselines, with ablation studies confirming complementary benefits from both components and robustness across shot counts and hyperparameter settings.

7 Acknowledgements

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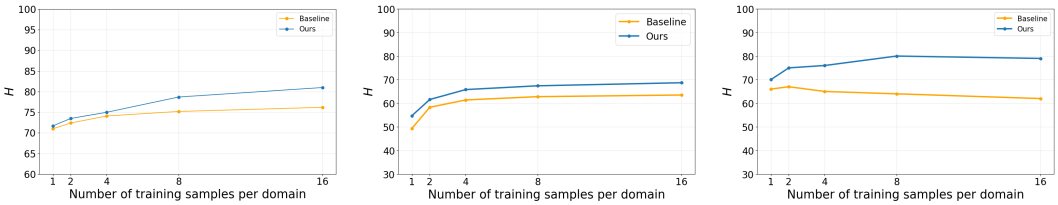
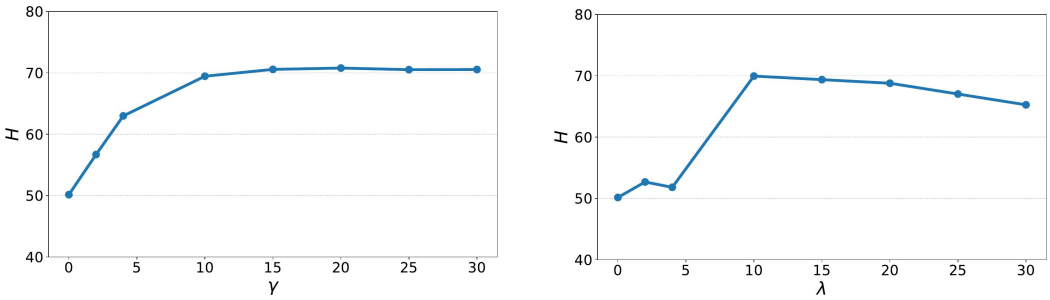


(a) ImageNet (b) Office-Home (c) Mini DomainNet

Figure 5: **Sensitivity to the Number of Training Samples per Domain.** We compare our method with the Baseline across varying numbers of shots. Our method consistently outperforms the Baseline across all datasets and shot counts.



(a) Impact of γ (b) Impact of λ

Figure 6: **Impact of Loss Weights γ and λ in Domain Disentangling Loss (DDL).** Performance remains stable across a range of hyperparameter values.

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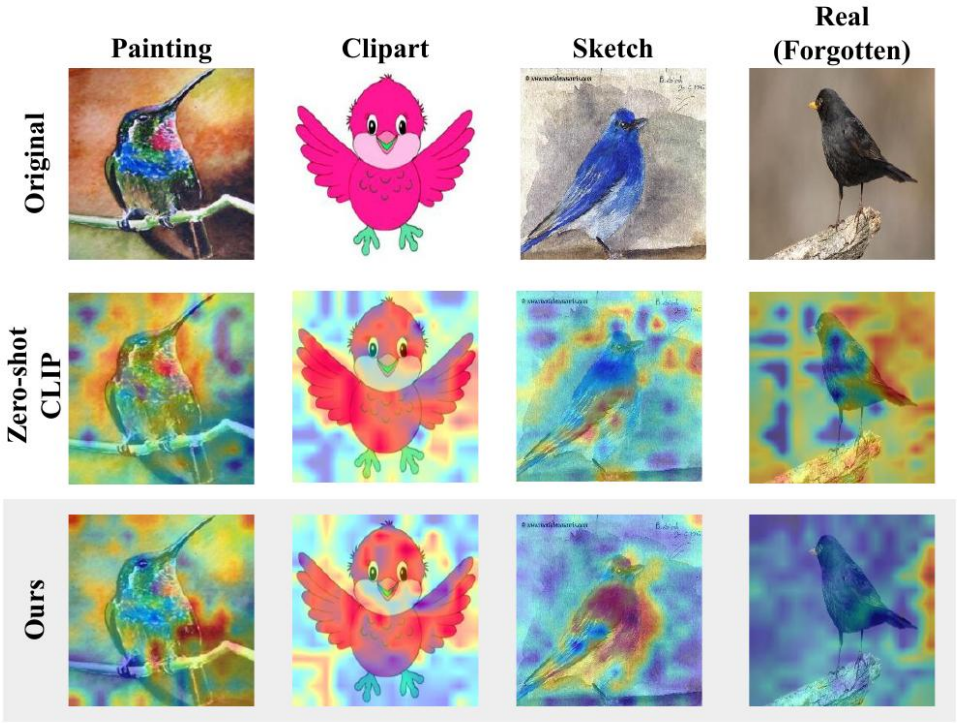

Figure 7: **Visualization of Attention Maps.** Our method attends to domain-relevant visual features, indicating that the learned representations capture meaningful domain-specific cues.

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